Clan Culture and the Gender Gap in Later-Life Health

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Abstract

Family institutions can insure members against hardship while assigning unequal rights and obligations to women and men. I study how this tension shapes later-life health by linking four waves of the China Health and Retirement Longitudinal Study to prefecture-level genealogy records. To isolate historical variation in clan intensity, I use distance to academies at which Zhu Xi taught and diffused Neo-Confucian principles of lineage organization. The 2SLS estimates imply that a one-standard-deviation increase in clan intensity narrows the gender gap in any limitation in activities of daily living by 3.8 percentage points, or about 16% of the outcome mean. Supporting analyses associate stronger clan culture with more children and sons, lower market-work intensity among women relative to men, and a larger grandparental-childcare response among older men. These patterns reveal a gendered allocation of work and family responsibilities, although they do not identify a causal mediation pathway. The findings connect historically rooted family organization to gender inequality in later-life physical health.

Keywords: Clan culture; Healthy aging; Gender inequality

JEL Codes: I14; J14; Z13

1 Introduction

Families are a central source of insurance and care where formal provision is incomplete, but the rules that sustain family support may also distribute work, care, and access to resources unequally. Chinese clans embody this tension. Kinship networks facilitate cooperation, risk sharing, and assistance during hardship, whereas patrilineal rules privilege male descendants and assign different obligations to women and men (Cao et al., 2022; Chen et al., 2022b; Greif and Tabellini, 2017; Zhang, 2022). Patrilineal continuity and son preference can encourage larger families and a division of labor in which men perform more market work while women assume more childbearing and domestic work (Zhang and Li, 2017; Zhang, 2022). Clan obligations can simultaneously provide informal insurance, elder support, and intergenerational care (Cao et al., 2022; Zhang, 2019). These forces have competing health implications: less market work may reduce occupational strain, additional births and unpaid care may increase total workload, and access to family support may differ from the obligation to provide it. I ask whether this combination of mutual support and gender hierarchy changes the gender gap in physical functioning later in life.

Health inequalities at older ages reflect exposures accumulated across the life course. Women in China live longer than men but spend fewer of their remaining years independently, revealing a functional-health disadvantage despite their longevity advantage (Chen et al., 2022a; Murtagh and Hubert, 2004; Ren et al., 2025). Employment histories, childbearing, unpaid work, and access to family care can all contribute to this divergence (Ervasti et al., 2021; Sun et al., 2025). Activities of daily living (ADLs) provide a direct measure of the resulting disadvantage: an ADL limitation records whether health problems interfere with dressing, bathing, eating, and other basic tasks, linking functional capacity to independent living and care needs (Chen et al., 2022a; Spector and Fleishman, 1998). This focus is increasingly important as China's population ages and demand for long-term care grows (Gong et al., 2022; Lei et al., 2022). It also shifts the question away from whether clans improve average health toward whether they change the distribution of healthy aging between women and men.

I study this question by combining the 2011, 2013, 2015, and 2018 waves of the China Health and Retirement Longitudinal Study (CHARLS) with prefecture-level genealogy records. The analysis covers 26,077 person-wave observations for Han respondents aged 60–90. I measure clan intensity as the natural logarithm of one plus genealogies compiled before 1990 per 10,000 residents and measure functional health with an indicator for any ADL limitation. The estimating equation interacts clan intensity with Female. The parameter of interest is therefore the difference between women's and men's clan-health gradients, not a separately identified effect for either group.

Genealogy density may reflect persistent regional conditions that also shape health. I address this concern by using the logarithm of distance to academies at which Zhu Xi taught and diffused

Neo-Confucian principles of clan organization (Chen et al., 2022b). Academy distance instruments Clan, while its interaction with Female instruments the corresponding gender differential. The controlled first stage has a Kleibergen–Paap rk F statistic of 29.47. The controlled 2SLS estimate implies that a one-standard-deviation increase in clan intensity narrows the gender gap in any ADL limitation by 3.8 percentage points, approximately 16% of the outcome mean. The implied slope for women is statistically indistinguishable from zero. The estimate thus identifies a gender difference in the clan gradient rather than a standalone health improvement for women.

The design is valid if academy proximity affects later-life health only through clan intensity, conditional on the controls. This exclusion restriction is the central identification concern because academy locations may be related to other persistent regional processes. I condition on predetermined geography, examine whether academy distance predicts development and health resources in 1990, and compare the result with alternative historical instruments. These exercises address observable threats but cannot prove the exclusion restriction.

The main pattern extends to chronic disease, hospitalization, and self-rated health and is unchanged after adding religious controls. It is also robust to restrictions on age, panel composition, and high-genealogy regions and to alternative constructions of clan intensity. Two instruments used in the recent clan literature yield estimates with the same sign, although their weaker first stages make them corroborating rather than decisive evidence (Ding et al., 2025; Fan et al., 2023).

Supporting analyses locate the result within a broader gendered allocation of family responsibilities. Stronger clan culture is associated with more children and sons and with lower employment and work intensity among women relative to men. A smaller CGSS exercise points in the same direction for domestic work, while CHARLS shows a larger increase in grandparental childcare among older men. I find little corresponding gender difference in social activity, exercise, sleep, drinking, or smoking. These outcomes are compatible with different lifetime allocations of paid work, unpaid work, and family support, but they do not identify a causal mediation pathway.

The paper contributes in three ways. First, it brings informal family organization into research on healthy aging and gender inequality in health. Existing studies show that medical access, social insurance, pensions, and long-term care can improve health or reduce dependence at older ages (Gruber et al., 2023; Huang and Zhang, 2021; Lei et al., 2022). Family structure and public policy also alter who supplies care, whether generations live together, and how older adults and their children share the burden of disability (Bau, 2021; Guo et al., 2025; Mommaerts, 2018). A related life-course literature documents that long working hours, childbirth, and the unequal division of paid and unpaid work can generate persistent health differences (Ervasti et al., 2021; Sun et al., 2025). I add a historically rooted determinant of these exposures. Clan organization can shape fertility, work, and access to support long before functional limitations emerge, connecting gender differences in independent living to an institution that predates modern health and social-insurance programs.

Second, I extend research on clans and gender inequality from economic behavior during working and reproductive ages to functional health later in life. Inherited gender norms continue to shape fertility and women's labor supply after migration and institutional change (Alesina and Giuliano, 2015; Alesina et al., 2013; Fernández and Fogli, 2009). In China, stronger lineage organization is associated with higher fertility and son preference, lower employment and income among women, and differences in women's social status (Huang et al., 2024; Zhang and Li, 2017; Zhang, 2022). The health prediction is less direct because lower market work, greater unpaid work, and stronger family support can have offsetting effects. The pooled gender contrast is also substantively useful: rather than asking whether people in clan-intensive places are healthier, it asks whether women and men exposed to the same local clan environment exhibit different health gradients. The paper thereby links the economic consequences of patrilineality during adulthood to a cumulative outcome that determines independence and care needs at older ages.

Third, the paper connects two views of clan organization that are often studied separately. Work on finance, disaster relief, local development, and family support emphasizes how clans mobilize resources, sustain cooperation, and protect members when markets or public programs are incomplete (Cao et al., 2022; Chen et al., 2022b; Ding et al., 2025; Fan et al., 2023; Zhang, 2019). Research on patrilineality instead emphasizes hierarchy, son preference, and unequal family rights and obligations. Mutual insurance and gender hierarchy, however, can be produced by the same lineage rules. The supporting outcomes place these functions in a common life-course framework: clan intensity predicts a more gendered allocation of fertility and paid work, stronger traditional gender-role attitudes, and a larger current childcare response among older men. These outcomes do not decompose the health coefficient or establish mediation. They show why treating the clan only as social capital or only as patriarchal restriction misses how its benefits and burdens are distributed within families.

Section 2 describes the institutional background. Section 3 presents the data and empirical strategy. Section 4 reports the main results and identification checks. Section 5 examines mechanisms, Section 6 considers heterogeneity and other health outcomes, and Section 7 concludes.

2 Institutional Background

Chinese clans are patrilineal kinship organizations whose members trace descent from a common male ancestor. Genealogies identify membership, locate individuals within a lineage hierarchy, preserve rules of conduct, and record obligations across generations. Together with ancestral halls, collective property, and recurring rituals, they provided an organizational infrastructure for exchanging information, disciplining behavior, pooling resources, and assisting relatives in need (Chen et al., 2022b; Fei, 1946; Greif and Tabellini, 2017; Peng, 2004). Collective property could

finance schooling, relief, and assistance to poorer members, while repeated interaction and lineage sanctions enforced transfers that were not written as market contracts. These arrangements lowered the cost of cooperation within the kinship boundary, but support depended on recognized membership and compliance with the duties attached to one's position. Clan capacity and clan hierarchy thus arose from the same organizational rules.

The geographic variation used below emerged over a long historical arc. Confucian family organization expanded under the Han dynasty, and Zhu Xi and other Song scholars later codified family rituals and helped diffuse genealogies and ancestral halls beyond elite households. By the Ming and Qing dynasties, clans had become important providers of local order, education, emergency relief, and support. Clan organizations were suppressed during the Maoist period, when many physical records were destroyed, especially during the Cultural Revolution, but lineage activities revived after the economic reforms that began in 1978 (Ebrey, 2014; Peng, 2004). This history motivates both the genealogy-based measure of clan intensity and the use of proximity to the academies at which Zhu Xi taught as a source of historical variation.

Mutual support operated within a hierarchical allocation of rights and duties. Claims on family resources and obligations to provide care depended on generation, age, and gender (Chen et al., 2022b). Male descendants occupied the central position in inheritance, ancestor worship, and continuation of the family line, making sons both heirs and expected providers of old-age support (Freedman, 2021; Oliveira, 2016). Marriage reinforced this asymmetry: a married woman generally entered her husband's lineage network, whereas sons remained responsible for their natal line. Women's standing was consequently tied to marriage, childbearing, and especially the production of male descendants. Market earnings, domestic production, childcare, and eldercare became different ways of fulfilling obligations to the extended family (Zhang and Li, 2017; Zhang, 2022).

These rules have competing implications for later-life health. Son preference and patrilineal continuity may increase fertility and shift women away from market employment and toward childbearing and domestic production, while men accumulate greater exposure to paid work. Yet unpaid work and caregiving can also be physically and psychologically costly, and unequal control over family resources may disadvantage women. Clan obligations further extend across generations: adult children support older parents, while grandparents provide childcare and other services to younger households (Sun et al., 2025; Zhang, 2019). For cohorts who spent most of adulthood before the expansion of formal pensions, health insurance, and market-based care, these arrangements can shape cumulative work, fertility, and support over the life cycle. The resulting gender difference in functional health cannot be signed from the institution alone.

Formal clan organizations no longer govern local society as they once did, but the associated norms and networks remain economically relevant. Contemporary evidence links clan strength to coordination, informal protection, fertility, household structure, and gendered economic outcomes

(Cao et al., 2022; Ding et al., 2025; Fan et al., 2023; Tang and Zhao, 2023; Wang et al., 2026). Genealogy density therefore measures exposure to a persistent local clan environment rather than individual membership or personal endorsement of lineage rules. The empirical analysis asks whether the health gradient associated with that environment differs by gender; the supporting outcomes then examine where gendered differences in fertility, work, care, and health behavior appear.

3 Data and Empirical Strategy

3.1 Data and Variables

The individual-level data come from the China Health and Retirement Longitudinal Study (CHARLS), a nationally representative survey of Chinese residents aged 45 and older (Zhao et al., 2013). I pool the 2011, 2013, 2015, and 2018 waves and link respondents to historical and geographic variables by prefecture of residence.[1] The baseline sample contains 26,077 person-wave observations for 9,092 Han respondents aged 60–90 in 5,954 households and 105 prefectures. The ethnicity restriction aligns the estimation sample with the patrilineal institution captured by the genealogy measure, while the age restriction focuses on a population for whom functional limitations are common without relying on the sparse extreme-age tail. Appendix Table B.1 documents the sample construction and panel coverage.

The primary outcome indicates whether a respondent has any limitation in activities of daily living, a standard measure of the ability to perform basic self-care independently (Chen et al., 2022a; Spector and Fleishman, 1998). CHARLS records limitations in dressing, bathing, eating, toileting, getting into or out of bed, and continence. The indicator equals one when at least one activity is limited, so coefficients are measured in percentage points. Chronic disease, hospitalization, and self-rated health provide complementary measures of morbidity, medical use, and overall health. The baseline controls are age and rural residence at the individual level and logged distance to the coastline and the five Qing treaty ports, river density, and mean terrain slope at the prefecture level.

I measure clan intensity using the Shanghai Library's *General Catalog of Chinese Genealogy* (*Zhongguo jiapu zongmu*).[2] Genealogies record male-line descent and family rules, and their compilation required collective effort to maintain a lineage. The stock of surviving books is

[1] I exclude the processed 2020 follow-up because it does not contain harmonized measures of social activity, employment status, and grandparental care used in the supporting analyses. Retaining the 2011–2018 waves keeps the baseline and supporting analyses on a common four-wave panel.

[2] See https://jiapu.library.sh.cn/. Genealogy density may reflect differential preservation and collection, and current residence need not coincide with lifetime exposure. Section 4 examines the observable implications of these limitations.

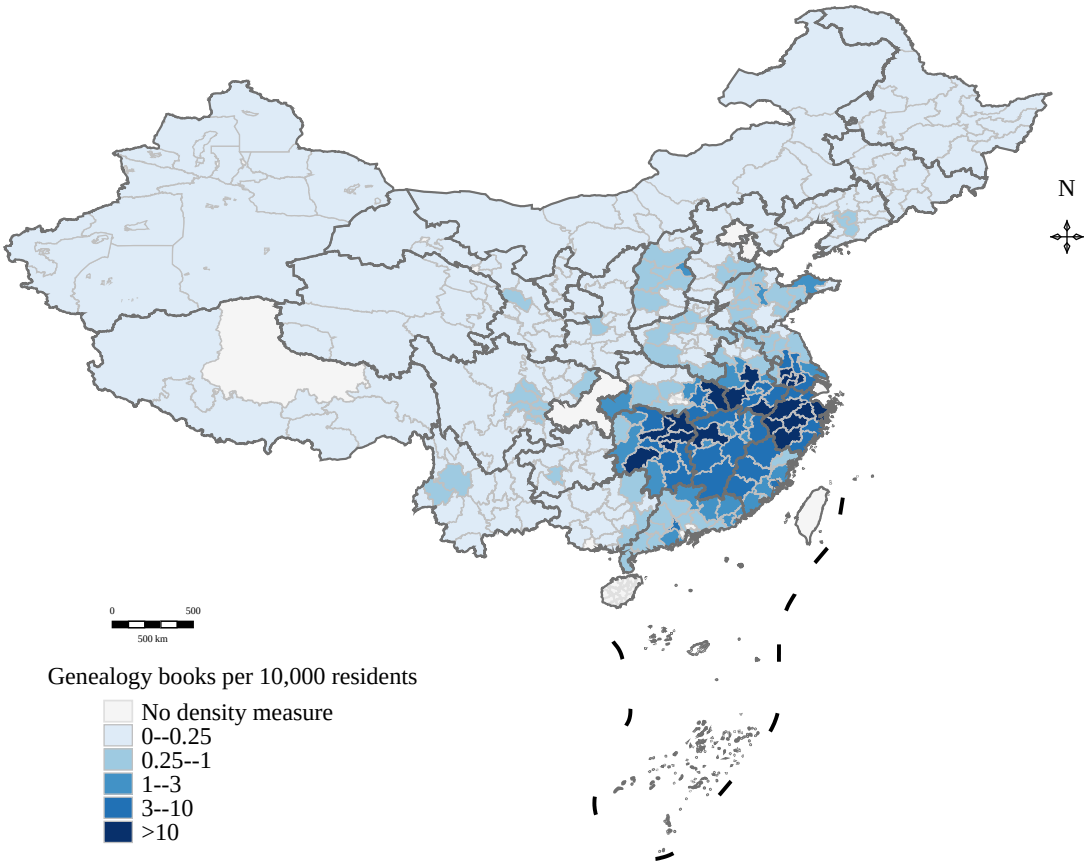


Figure 1. Genealogy Density Across Prefectures

Note: The map reports genealogy books per 10,000 residents for all 323 prefectures with a nonmissing density measure in the genealogy data. It is not restricted to the CHARLS estimation sample. Gray areas lack a genealogy-density observation. The administrative-boundary file has map approval number GS(2024)0650.

therefore a prefecture-level proxy for historical clan organization (Chen et al., 2022b; Greif and Tabellini, 2010, 2017). Of 52,680 catalog records across 346 prefectures, 48,658 contain usable location and compilation information. I retain books compiled before 1990, aggregate them by prefecture, and divide by population in the 1990 Population Census. Clan is the natural logarithm of one plus the resulting number of books per 10,000 residents. This normalization separates lineage intensity from prefecture size, retains prefectures with no recorded genealogy, and limits the leverage of the right tail.

The supporting analyses use additional CHARLS outcomes on fertility, employment, grandparental care, cognition, mental health, social activity, and lifestyle. The Chinese General Social Survey (CGSS) supplies housework and gender-role measures that CHARLS does not record consistently, and the 1990 Population Census provides an earlier cross-section on fertility and son composition. These samples are analyzed separately and provide converging evidence rather than observations in a single causal chain. Appendix Table B.2 lists each source and survey year; Table

1 reports summary statistics for the baseline variables.

Figure 1 maps genealogy density for all prefectures with usable records, not only those sampled by CHARLS. The measure is concentrated in a southeastern and south-central belt but is not confined to Fujian and Guangdong. Appendix Figure A.2 shows the distribution of compilation years.

Panel A of Table 1 summarizes the individual and household variables in the estimation sample. Functional limitations are common but not universal: 23% of person-wave observations report at least one ADL limitation. Chronic conditions are more prevalent, affecting 81% of observations, while 17% report a hospitalization. Respondents are 68.2 years old on average; 62% live in rural areas, 81% are married, and 95% report health-insurance coverage. The smaller observation counts for exercise, consumption, financial transfers, and several health measures reflect item- and wave-specific nonresponse, so analyses using these outcomes rely on their respective nonmissing samples.

Panel B shows the distribution of clan intensity and the predetermined prefecture-level geographic controls. Clan intensity varies substantially across the 105 baseline prefectures: the untransformed genealogy measure averages 2.57 books per 10,000 residents, has a standard deviation of 5.93, and ranges from zero to 35.48. This dispersion, together with the long right tail, motivates the logarithmic transformation used in the regressions. The average prefecture lies 421.92 kilometers from the coastline and 417.66 kilometers from the nearest Qing treaty port, while river density and terrain slope capture local transport, agricultural, and accessibility conditions. These variables account for geographic factors that may have jointly shaped historical clan formation and long-run economic development; they do not exhaust all regional differences that may also matter for health. The summary statistics establish the composition and support of the estimation sample but do not by themselves identify a gender-differentiated relationship between clan intensity and health.

3.2 Descriptive Evidence

I first describe how functional limitations vary with clan intensity before introducing the instrumental-variable design. The fitted lines in Figure 2 correspond to the descriptive regressions

$$\bar{y}_{pg} = \alpha_g + \beta_g Clan_p + \varepsilon_{pg}, \tag{1}$$

where $\bar{y}_{pg}$ is the prefecture-level mean ADL-limitation rate for gender g, and $Clan_p$ is logged genealogy density. These regressions summarize the raw cross-prefecture relationship and do not include the controls used in the baseline specification.

Figure 2 plots prefecture-level mean ADL limitations against clan intensity separately for

Table 1. Summary Statistics

Variables	Obs.	Min	Mean	Max	SD
Panel A: Individual and household variables (CHARLS)					
Any ADL limitation	26,077	0.00	0.23	1.00	–
Any chronic condition	25,675	0.00	0.81	1.00	–
Hospitalization	26,034	0.00	0.17	1.00	–
Health insurance	25,600	0.00	0.95	1.00	–
Social activity (1=yes)	26,077	0.00	0.49	1.00	–
Exercise (1=yes)	15,976	0.00	0.87	1.00	–
Nap (1=yes)	25,144	0.00	0.61	1.00	–
Work status (1=yes)	26,077	0.00	0.10	1.00	–
Marital status	26,077	0.00	0.81	1.00	–
Rural (1=yes)	26,077	0.00	0.62	1.00	–
Caring for grandchildren (1=yes)	26,075	0.00	0.36	1.00	–
Self-rated health	24,862	1.00	2.97	5.00	0.95
CES-D (score 0–30)	24,207	0.00	8.32	30.00	6.29
Sleep time (hours)	24,906	0.00	6.17	24.00	2.06
Age	26,077	60.00	68.20	90.00	6.59
Education	25,458	1.00	1.74	4.00	0.96
Family size	26,077	1.00	2.98	16.00	1.66
Consumption per capita (1,000 yuan)	19,030	0.00	12.27	1,006.91	20.58
Financial support to children (1,000 yuan)	25,132	0.00	2.53	1,003.00	20.65
Number of children	26,077	0.00	3.12	10.00	1.48
Number of sons	26,077	0.00	1.63	7.00	1.07
Panel B: Prefecture variables					
Genealogies per 10,000 residents	105	0.00	2.57	35.48	5.93
Distance to coastline (km)	105	14.38	421.92	1,369.29	361.18
Distance to treaty port (km)	105	11.25	417.66	1,388.75	362.43
River density	105	0.06	0.25	1.40	0.18
Slope	105	1.60	10.38	23.00	5.48

Note: Panel A reports nonmissing observations from the 2011, 2013, 2015, and 2018 CHARLS waves in the baseline sample. Panel B reports one observation per prefecture. Consumption and financial support are measured in 1,000 yuan. Educational attainment equals 1 for below primary school, 2 for primary school, 3 for middle school, and 4 for high school or above. Genealogy density is the number of books compiled before 1990 per 10,000 residents; the regressions use its natural logarithm after adding one. Geographic distances are measured from each prefecture's geographic center and reported in kilometers; the regressions use their natural logarithms. River density is total river length divided by administrative area.

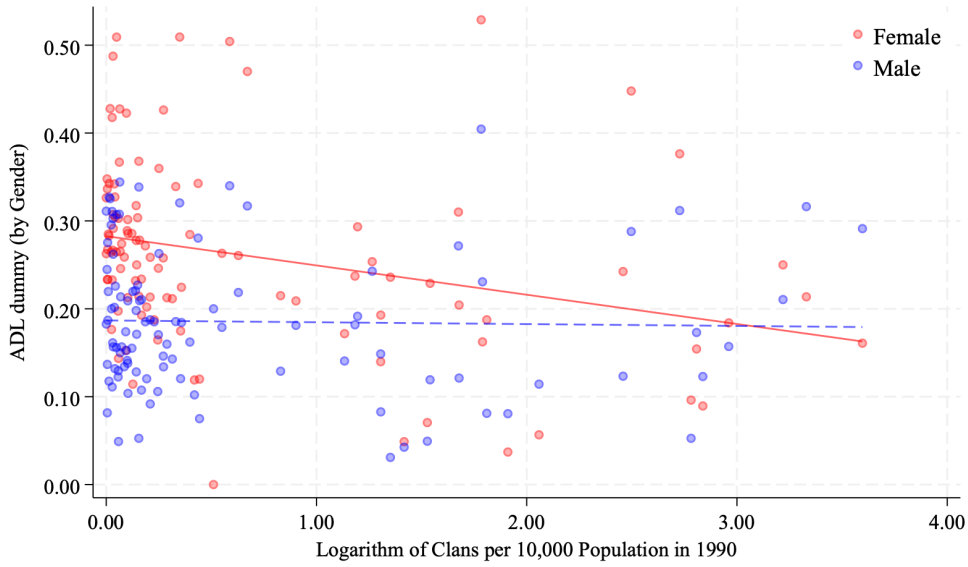


Figure 2. Clan Intensity and ADL Limitations

Note: The figure plots prefecture-level mean ADL-limitation rates against clan intensity separately for women and men. The solid line represents women and the dashed line represents men. Clan intensity is the natural logarithm of one plus genealogy books compiled before 1990 per 10,000 residents.

women and men. Women have higher limitation rates on average, but their fitted slope is more negative.[3] This pattern motivates the pooled interaction specification below. It remains purely descriptive: the fitted lines neither isolate a causal effect nor establish that the two slopes differ. Appendix Figure A.1 presents analogous plots for the alternative health outcomes.

3.3 Empirical Strategy

The baseline specification estimates whether the association between clan intensity and ADL limitations differs by gender. I pool person-wave observations and include wave fixed effects, but not individual fixed effects: Clan is measured at the prefecture level and is effectively time invariant for respondents who remain in the same prefecture. Identification therefore comes from cross-prefecture variation in clan intensity and its interaction with gender. I estimate

$$y_{ipt} = \beta_0 + \beta_1 Clan_p + \beta_2 Clan_p \times Female_i + \beta_3 Female_i + \gamma_1 X_p + \gamma_2 X_{ip} + \eta_t + \varepsilon_{ipt}. \tag{2}$$

where y_{ipt} is the ADL-limitation indicator for individual i in prefecture p and wave t, X_p contains the prefecture-level geographic controls, X_{ip} contains age and rural residence, and η_t denotes

[3]Older women have lower mortality and lower prevalence of some chronic diseases but report more functional limitations and disability than older men (Murtagh and Hubert, 2004; Nathanson, 1975; Ren et al., 2025).

wave fixed effects. The coefficient of interest, β_2, is the difference between the female and male clan gradients. The male slope is β_1, and the female slope is $\beta_1 + \beta_2$. The coefficient β_3 measures the gender difference when Clan equals zero rather than the average gender gap in the sample. Standard errors are clustered by prefecture, the level at which Clan, academy distance, and the geographic controls vary. Repeated person-wave observations improve the measurement of health and the gender contrast but do not create additional independent variation in clan intensity.

OLS estimates may reflect unobserved prefecture characteristics, including persistent differences in social capital, economic development, and health inputs (Guiso et al., 2006). I therefore instrument clan intensity with the natural logarithm of the shortest distance from each prefecture to an academy at which Zhu Xi taught (Chen et al., 2022b; Fan et al., 2023).

Zhu Xi helped popularize Confucian practices related to clan organization from the twelfth century onward (Chen et al., 2022b). He taught at Yuelu Academy in Changsha, Hanquan Academy in Jianyang, and Bailudong Academy in Jiujiang, and his *Rituals of the Family* supplied practical rules for lineage organization that spread widely from the thirteenth to the nineteenth century (Ebrey, 2014). With high historical transportation and communication costs, proximity facilitated this diffusion. The design requires academy distance to affect later-life health only through clan intensity after conditioning on the controls. Section 4 discusses threats to this exclusion restriction and the available diagnostic evidence. The two first-stage equations and the 2SLS equation are

$$Clan_p = \pi_0 + \pi_1 lnDis_p + \pi_2 lnDis_p \times Female_i + \pi_3 Female_i + \gamma_1 X_p + \gamma_2 X_{ip} + \eta_t + u_{ipt}. \quad (3)$$

$$Clan_p \times Female_i = \rho_0 + \rho_1 lnDis_p + \rho_2 lnDis_p \times Female_i + \rho_3 Female_i + \gamma_1 X_p + \gamma_2 X_{ip} + \eta_t + v_{ipt}. \quad (4)$$

$$y_{ipt} = \beta_0 + \beta_1 \widehat{Clan_p} + \beta_2 (\widehat{Clan_p \times Female_i}) + \beta_3 Female_i + \gamma_1 X_p + \gamma_2 X_{ip} + \eta_t + \varepsilon_{ipt}. \quad (5)$$

The variables follow equation (2). Because both $Clan_p$ and $Clan_p \times Female_i$ are treated as endogenous, the excluded instruments are $lnDis_p$ and $lnDis_p \times Female_i$. The coefficient β_2 in equation (5) captures the difference between the female and male clan slopes.

The two endogenous regressors and two excluded instruments make this a just-identified system. The reported Kleibergen–Paap statistic evaluates the joint relevance of the excluded instruments under prefecture-level clustering. The 2SLS coefficient is identified by the component of clan intensity associated with proximity to the academies at which Zhu Xi taught, so it need not equal the average relationship generated by every source of variation in contemporary clan culture. This source of variation and the exclusion restriction should be kept in view when comparing OLS and IV estimates.

4 Empirical Results

4.1 Baseline Results

Clan intensity is associated with a smaller female disadvantage in functional health. In Table 2, the controlled OLS estimate of the gender difference in the clan gradient is -0.034, and the corresponding 2SLS estimate is -0.044. This difference is negative before instrumenting and remains stable when the baseline controls are added. The larger IV magnitude may reflect measurement error, omitted regional factors, or the particular variation in clan intensity induced by academy proximity; the OLS–IV comparison cannot distinguish among these explanations.

The magnitude implies a meaningful compression of the gender gap. A one-standard-deviation increase in logged clan intensity narrows women's disadvantage in ADL limitations by 3.8 percentage points, approximately 16% of the outcome mean. The estimated clan slope is 0.024 for men and -0.020 for women, although the latter is statistically indistinguishable from zero. The implied gender gap falls from 0.113 at the 10th percentile of Clan to 0.035 at the 90th percentile. It becomes slightly negative only at the 95th percentile, where the estimate is too imprecise to establish a reversal. The evidence therefore identifies a differential slope and a narrowing of women's functional-health disadvantage, not a general health benefit of clan culture or a separately identified improvement for women.

The first stage supports the relevance of academy distance. In the controlled specification, greater distance predicts lower clan intensity: the coefficient is -0.786 with a standard error of 0.102, and the Kleibergen–Paap rk F statistic is 29.47 (Kleibergen and Paap, 2006). The sign accords with the historical diffusion argument, but instrument relevance does not establish the exclusion restriction. Moreover, the instrument varies across 105 prefectures rather than across 26,077 person-wave observations. The micro data sharpen measurement of individual health and the gender contrast; the independent identifying variation remains geographic, as reflected in prefecture-clustered standard errors.

4.2 Robustness and Identification Checks

The baseline estimate could depend on how health and clan intensity are measured, the historical source of identifying variation, or the geographic and longitudinal composition of the sample. The following exercises address each concern in turn.

4.2.1 Alternative Outcomes and Additional Controls

The binary ADL indicator captures whether respondents report any difficulty with basic activities, but it may not reflect broader dimensions of health. I therefore consider three outcomes drawn

Table 2. Baseline Results

	Any ADL limitation			
	OLS		IV	
	(1)	(2)	(3)	(4)
			First stage for Clan	
Log academy distance			-0.823***	-0.786***
			(0.114)	(0.102)
KP rk F			26.11	29.47
			2SLS	
Clan	0.001	0.010	0.017	0.024*
	(0.010)	(0.009)	(0.012)	(0.012)
Female	0.108***	0.108***	0.114***	0.114***
	(0.010)	(0.009)	(0.011)	(0.011)
Clan × Female	-0.034***	-0.034***	-0.043***	-0.044***
	(0.009)	(0.009)	(0.012)	(0.012)
Controls	NO	YES	NO	YES
Wave FE	YES	YES	YES	YES
Obs.	26,077	26,077	26,077	26,077
Adj. R^2	0.015	0.054	0.015	0.053

Note: The dependent variable is the indicator for any ADL limitation. Columns (1) and (2) report OLS estimates; columns (3) and (4) report 2SLS estimates. The displayed first-stage coefficient comes from the equation for Clan. Logged academy distance and its interaction with Female instrument for Clan and Clan × Female, respectively; the interaction instrument is included in estimation but omitted from the table. All specifications include wave fixed effects; columns (2) and (4) add the baseline controls. Standard errors clustered at the prefecture level in parentheses. Data are from CHARLS. KP rk F is the Kleibergen–Paap Wald rk F statistic (Kleibergen and Paap, 2006). * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

from CHARLS. Chronic disease indicates whether a respondent reports at least one of 14 doctor-diagnosed conditions; hospitalization records admission to a hospital during the previous year; and self-rated health uses a five-point scale from very poor to very good. The first two outcomes capture morbidity and recent inpatient use, whereas the third summarizes the respondent's broader assessment of health. These variables are not clinical records, but their different recall windows and measurement scales reveal whether the baseline pattern is confined to functional limitations.

The alternative outcomes preserve the same gender-differentiated pattern. In columns (2)–(4) of Table 3, stronger clan intensity is associated with lower chronic-disease and hospitalization rates for women relative to men and with better self-rated health. The estimates show that the baseline pattern is not confined to the coding of ADL limitations; they should not be interpreted as separate causal effects on disease incidence, medical utilization, or health perceptions.

Column (1) instead retains the ADL outcome and adds prefecture-level counts of Buddhist and Taoist religious sites. The counts are logged after adding one and proxy for the contemporaneous local religious environment.[4] Religious institutions may be correlated with both local cultural organization and health-related behavior. The interaction estimate is unchanged from the controlled baseline, which indicates that the result is not mechanically attributable to this particular measure of local religious presence.

4.2.2 Alternative Instruments

The main IV strategy relies on distance to the academies at which Zhu Xi taught, so a natural concern is that the result reflects other long-run consequences of academy geography rather than clan organization. Table 4 uses two alternative sources of historical variation: distance to prefectures associated with the southward migration of 1127–1130 and the local rice–wheat suitability ratio (Ding et al., 2025; Fan et al., 2023; Talhelm et al., 2014; Wang et al., 2026). The first reflects migration-generated competition among lineages, whereas the second captures agricultural conditions associated with coordination and social organization. Both produce negative estimates of the female–male difference in the clan gradient, with magnitudes close to the baseline estimate. Their Kleibergen–Paap statistics are lower than in the academy-distance design, and each instrument carries its own exclusion restriction. They therefore provide corroborating evidence rather than a substitute for the principal instrument.

The exclusion restriction for academy distance remains the central identifying concern. Academy proximity may have affected later-life health through educational diffusion, economic

[4]The underlying register was downloaded from the National Religious Affairs Administration's searchable registry of religious activity venues (http://www.sara.gov.cn/zjhdcsjbxx/index.jhtml). The source code downloads the registry, geocodes the recorded addresses, spatially matches sites to 2020 county boundaries, and aggregates Buddhist and Taoist sites to the prefecture. It is a cross-sectional register of recorded sites rather than a historical measure of religious affiliation or practice.

Table 3. Alternative Outcomes and Additional Controls

	Religion controls	Any chronic condition	Hospitalization	Self-rated health
	(1)	(2)	(3)	(4)
Clan × Female	-0.044***	-0.038***	-0.028***	0.071***
	(0.012)	(0.011)	(0.009)	(0.022)
Controls	YES	YES	YES	YES
Wave FE	YES	YES	YES	YES
Obs.	26,077	25,675	26,034	24,862
Adj. R^2	0.053	0.032	0.022	0.030
KP rk F	29.47	29.74	29.50	29.86

Note: Column (1) adds the logarithms of one plus the prefecture-level numbers of registered Buddhist and Taoist sites. Columns (2)–(4) use, respectively, an indicator for any of 14 reported chronic conditions, an indicator for hospitalization during the previous year, and self-rated health on a five-point scale from very poor to very good. The table reports Clan × Female; the main effects are included in estimation but omitted. Each specification instruments Clan and Clan × Female with academy distance and its interaction with Female, includes the baseline controls and wave fixed effects, and clusters standard errors at the prefecture level. Health outcomes are from CHARLS; the religious-site register is described in the text. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

development, health infrastructure, or gender norms independently of clan organization. The baseline specifications condition on predetermined geography—distance to the coastline and Qing treaty ports, river density, and terrain slope. Appendix Table B.3 then relates academy distance to GDP, grain output, secondary schools, and physicians in 1990 for the 94 prefectures with complete *China City Statistical Yearbook* data. Conditional on the geographic controls, none of these associations is detectable. This diagnostic addresses a limited set of observable long-run correlates; it cannot test the exclusion restriction. The IV estimates should therefore be interpreted as the gender-differentiated effect of the component of clan intensity predicted by historical academy proximity.

4.2.3 Other Robustness Checks

Figure 3 combines checks addressing four remaining concerns. First, health and continued survey participation both vary with age, so the composition of an older longitudinal sample can change as respondents die or leave the survey. Panel attrition can distort estimates when continued observation is related to outcomes or their determinants, a concern documented both in general household panels and in aging studies such as the English Longitudinal Study of Ageing and the U.S. Health and Retirement Study (Banks et al., 2011; Fitzgerald et al., 1998). Restricting the sample to ages 60–75 reduces the influence of respondents at the highest mortality risk, while the four-wave balanced sample holds the set of observed respondents fixed across waves. Neither

Table 4. Alternative Instruments

	Migration distance		Rice–wheat ratio	
	First stage	2SLS	First stage	2SLS
	(1)	(2)	(3)	(4)
Log migration distance	-0.565***			
	(0.133)			
Rice–wheat ratio			0.914***	
			(0.181)	
Clan × Female		-0.038**		-0.041***
		(0.018)		(0.014)
Controls	YES	YES	YES	YES
Wave FE	YES	YES	YES	YES
Obs.	26,077	26,077	26,077	26,077
Adj. R^2	0.311	0.054	0.403	0.054
KP rk F	-	9.04	-	12.75

Note: This table replaces academy distance with distance to the nearest prefecture associated with the 1127–1130 southward migration and with the rice–wheat suitability ratio. The 2SLS columns report the coefficient of interest, Clan × Female; the main effects are included in estimation but omitted from the table. All specifications include the baseline controls and wave fixed effects. Standard errors clustered at the prefecture level in parentheses. The migration-distance and rice–wheat instruments are supporting alternatives and rely on their own exclusion restrictions. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

restriction recovers outcomes after death or permanent attrition; they instead show whether the baseline estimate is unusually dependent on the composition of the observed sample.[5]

Second, the baseline coefficient compares women and men but does not require that they be observed with their own spouses. I therefore re-estimate the model in household-waves in which both married partners are observed.[6] It assesses whether the result depends on comparing women and men drawn from different observed household settings.

Third, historical and anthropological accounts place especially dense lineage organization in southern and southeastern China, including Fujian, while the Shanghai Library catalog displays a similar geographic concentration (Chen et al., 2022b; Freedman, 2021; Szonyi, 2002; Zheng, 2001). If a small set of high-density regions drives the identifying variation, the interaction could partly reflect their distinct economic development, migration histories, health systems, or other cultural institutions. I therefore exclude the academy provinces and, separately, the three provinces with the highest genealogy density.[7] These restrictions ask whether the baseline pattern extends beyond the regions in which clan institutions and the historical instrument are most concentrated.

Fourth, genealogy density is a proxy constructed from surviving historical books rather than a direct measure of individual clan membership. Research using the contents of Chinese genealogies emphasizes that these sources can reflect selective recording, retrospective updating, and survivor bias (Shiue, 2017). The measure uses book counts rather than individual entries, but it faces a related preservation problem: the catalog is strongly right-skewed, and books may have disappeared or been destroyed before collection (Chen et al., 2022b). I therefore change the lower-tail transformation and test whether zero records or coastal regions drive the result.[8] Together, these specifications assess whether the baseline estimate depends on a particular treatment of the catalog rather than the broader spatial variation in clan intensity.

Across these exercises, the estimated interaction remains negative and similar in magnitude. The baseline finding is therefore not tied to a single health outcome, instrument, sample restriction, region, or construction of clan intensity.

[5]The age restriction retains respondents aged 60–75. The balanced-panel specification retains respondents who satisfy the baseline restrictions and are observed in all four 2011–2018 waves.

[6]Operationally, this restriction retains household-waves containing exactly one married male and one married female respondent who both satisfy the baseline sample restrictions. It contains 7,968 household-waves (15,936 person-wave observations), equally divided between women and men. The harmonized CHARLS file does not retain a direct spouse identifier, so this is a conservative co-resident married-partner proxy rather than a confirmed respondent-level couple match.

[7]The academy-province exclusion removes Hunan, Fujian, and Jiangxi (17 of 105 baseline prefectures). The high-density exclusion removes Zhejiang, Hunan, and Jiangsu (19 prefectures), ranked using genealogy density calculated from all raw prefecture records.

[8]The IHS specification replaces $\ln(1 + \text{genealogy density})$ with $\text{asinh}(\text{genealogy density})$. "Positive genealogy density" excludes the two of 105 baseline prefectures with zero catalogued genealogy density. "Beyond 100 km from coast" retains 77 prefectures and excludes the 28 whose centers are at most 100 kilometers from the coast.

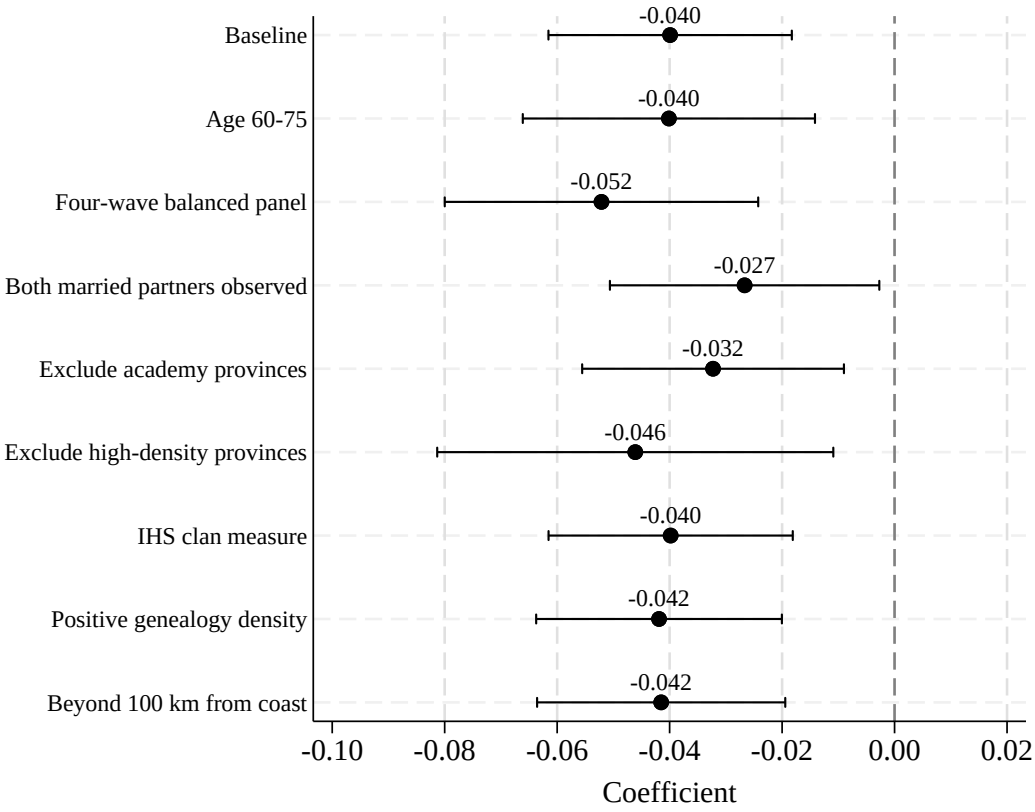


Figure 3. Sensitivity to Alternative Specifications

Note: Points report the estimated coefficient on Clan × Female, and horizontal bars show 95% confidence intervals based on standard errors clustered at the prefecture level. The dependent variable is any ADL limitation. Clan is standardized using the unweighted prefecture distribution in the full baseline sample, so coefficients represent a one-standard-deviation increase in the corresponding clan measure. All specifications include the baseline controls and wave fixed effects.

5 Mechanisms

The health estimates do not reveal when the gender difference develops or which family decisions produce it. Because culture can shape behavior across generations, I examine outcomes that capture paid work, unpaid work, and family support over the life course (Guiso et al., 2006, 2016).[9] The evidence is organized around fertility and labor supply, family responsibilities, and health-related behavior. These outcomes can support or weaken particular interpretations, but they do not constitute a formal mediation analysis.

[9]Using historical female labor-force participation and fertility in countries of ancestry as proxies for culture, Fernández and Fogli (2009) show that inherited norms predict women's fertility and labor supply.

5.1 Fertility and Labor Supply

Patrilineal continuity connects fertility decisions to the household division of labor. Male descendants traditionally carry the lineage, inherit family property, and are expected to support parents in old age (Freedman, 2021; Zhang, 2019). The resulting preference for sons can induce repeated childbearing (Basu and De Jong, 2010). Because women bear much of the cost of childbirth and childcare, the same norm can reduce their employment and work intensity even when it increases the family's desired number of children (Zhang and Li, 2017; Zhang, 2022).

Table 5. Children and Sons

	Aged 60–90		Aged 45–90		Aged 45–59	
	Children	Sons	Children	Sons	Children	Sons
	(1)	(2)	(3)	(4)	(5)	(6)
Clan	0.268***	0.159***	0.214***	0.127***	0.153**	0.091*
	(0.091)	(0.056)	(0.080)	(0.049)	(0.075)	(0.047)
Controls	YES	YES	YES	YES	YES	YES
Wave FE	YES	YES	YES	YES	YES	YES
Obs.	26,077	26,077	51,511	51,511	25,434	25,434
Adj. R^2	0.015	0.014	0.007	0.005	0.009	0.003
KP rk F	63.69	63.69	62.41	62.41	59.41	59.41

Note: This table reports 2SLS estimates for the numbers of children and sons in three age groups. Academy distance instruments Clan. All specifications include the baseline controls and survey-wave fixed effects. Standard errors clustered at the prefecture level in parentheses. Data are from CHARLS. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

The estimates follow this institutional logic. Among respondents aged 60–90, a one-unit increase in Clan is associated with 0.268 more children and 0.159 more sons (Table 5). The corresponding estimates fall to 0.153 and 0.091 among respondents aged 45–59. Younger cohorts spent more of their reproductive years under fertility restrictions introduced in the 1970s, which may have constrained the expression of clan-based fertility preferences.[10] The 1990 Population Census shows a similar positive relationship between clan intensity and fertility (Appendix Table B.4). Children and sons capture related but distinct implications of patrilineality. A higher number of sons can result from a larger family even without changes in the sex composition of births, while a son preference may also alter stopping behavior or sex selection. The estimates establish that stronger clan areas have both more children and more sons; they do not by themselves identify which reproductive decision generated the difference. The smaller coefficients for younger respondents and the census corroboration nevertheless make the fertility pattern difficult to

[10]The cohort pattern is descriptive because age groups differ along dimensions other than exposure to fertility policy (Zhang, 2017).

attribute to a single CHARLS wave or only to recall among the oldest respondents.

Paid work moves in the opposite direction for women relative to men. Table 6 focuses on respondents below age 60. Relative to men, women in areas with stronger clan culture are 5.4 percentage points less likely to work, work 0.224 fewer days per week, and work 0.312 fewer hours per day.[11] These estimates point to a more gendered allocation of paid work as clan intensity rises.

Table 6. Labor Supply

	Work status (1)	Workdays per week (2)	Daily work hours (3)
Clan	0.074***	0.262**	0.367**
	(0.022)	(0.108)	(0.158)
Female	-0.168***	-0.947***	-1.468***
	(0.012)	(0.078)	(0.118)
Clan × Female	-0.054***	-0.224***	-0.312***
	(0.013)	(0.077)	(0.120)
Controls	YES	YES	YES
Wave FE	YES	YES	YES
Obs.	26,745	24,007	24,012
Adj. R^2	0.205	0.178	0.172
KP rk F	30.06	29.22	29.20

Note: The dependent variables indicate current employment, workdays per week, and daily work hours. Nonworkers are coded as zero for the two intensity measures. The sample consists of Han CHARLS respondents below age 60. Academy distance and its interaction with Female instrument Clan and Clan × Female. All specifications include the baseline controls and wave fixed effects. Standard errors clustered at the prefecture level in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

The three labor outcomes distinguish participation from work intensity. The employment estimate shows a wider gender gap in participation as Clan rises, and the workday and work-hour estimates reveal the same pattern at the intensive margin. The measures cannot recover lifetime occupational exposure or distinguish physically demanding from sedentary jobs. They nevertheless document a behavioral margin through which women and men in the same clan environment may accumulate different histories of paid work.

Taken together, higher fertility and a wider gender difference in paid work point toward unequal cumulative physical demands over adulthood. The estimates do not show that reduced market work improves women's health: they do not measure total work, and fertility and labor supply are not randomly assigned mediators.

[11]The statutory retirement age is 60 for men, 50 for most women, and 55 for female cadres; rural China has no single mandatory retirement age.

5.2 Family Responsibilities

Less market work need not imply less total work if time shifts toward housework and care (Sun et al., 2025). CHARLS does not measure domestic work consistently, so Appendix Table B.5 uses Han respondents aged 45–59 in the 2015 Chinese General Social Survey. The self-reported interactions are positive but do not differ detectably from zero, whereas the spouse-reported interactions are negative and larger in magnitude. The estimates are based on a much smaller sample and the two reports are not directly comparable, so I treat them as supplementary evidence rather than a central result. The reporting structure of the CGSS also complicates interpretation. Respondents report their own time and their spouse's time, so the spouse outcome changes both the person whose work is measured and the gender of the reporter. The larger spouse-reported interaction may capture differences in knowledge or reporting as well as a larger division of domestic labor. For this reason, the appendix evidence establishes at most a directional pattern and is not used to quantify total work or explain the baseline coefficient.

Intergenerational care provides stronger evidence that clan obligations remain behaviorally relevant. In the samples used in Table 7, 36.3% of respondents cared for grandchildren in the previous year, the mean total across adult children's families is 16.0 caregiving weeks, and caregivers and noncaregivers together average 21.8 weekly hours under the survey measure. Columns (1), (3), and (5) associate a one-unit increase in Clan with a 3.1 percentage point rise in caregiving participation, 1.93 additional caregiving weeks, and 5.14 additional weekly hours. The gender-interacted specifications locate most of this response among older men. Columns (2), (4), and (6) imply male slopes of 4.4 percentage points, 2.51 weeks, and 6.22 hours; the negative interaction estimates indicate smaller responses among women. Relative to the corresponding outcome means, the male slopes amount to approximately 12%, 16%, and 29%. This pattern suggests that clan obligations reallocate current care toward older men, but it does not establish that grandparental childcare mediates the health relationship. Healthier older people may be more able to provide care, and current caregiving occurs after much of the relevant lifetime health exposure.[12]

The observed allocation of family responsibilities is accompanied by a similar pattern in stated gender roles. Table 8 uses the 2010–2015 CGSS and shows that older respondents in higher-clan areas express stronger agreement that men should prioritize careers while women should prioritize family, that men are more capable, and that marrying well is more important for women than working well. The estimates are similar when the sample is divided between older women and older men. Clan intensity is not detectably related to the statements about laying off women first

[12]The same family organization may also have provided these women with greater grandparental assistance earlier in adulthood, reducing their own childcare burden. This interpretation is conjectural: CHARLS records care currently supplied by older respondents rather than care they received while raising their children, so the historical support channel cannot be tested directly.

Table 7. Grandparental Childcare

	Cared last year		Weeks last year		Hours per week	
	(1)	(2)	(3)	(4)	(5)	(6)
Clan	0.031**	0.044***	1.928*	2.508**	5.137**	6.215**
	(0.013)	(0.013)	(1.010)	(1.037)	(2.443)	(2.661)
Female		-0.013*		0.052		0.607
		(0.007)		(0.340)		(1.818)
Clan × Female		-0.027***		-1.154***		-2.131*
		(0.008)		(0.434)		(1.190)
Controls	YES	YES	YES	YES	YES	YES
Wave FE	YES	YES	YES	YES	YES	YES
Mean of Dep.	0.363	0.363	16.000	16.000	21.798	21.798
Obs.	26,075	26,075	26,077	26,077	26,077	26,077
Adj. R^2	0.117	0.118	0.066	0.067	0.003	0.003
KP rk F	63.72	29.47	63.72	29.47	63.72	29.47

Note: The dependent variables indicate whether the respondent cared for grandchildren in the past year, the total weeks spent caring for grandchildren during that year, and the corresponding weekly hours. Respondents without eligible grandchildren are coded as zero. Odd-numbered columns estimate the average clan association; even-numbered columns allow it to differ by gender. Academy distance instruments Clan in odd-numbered columns; academy distance and its interaction with Female instrument Clan and Clan × Female in even-numbered columns. The mean of the dependent variable is calculated over each column's estimation sample. The sample consists of Han CHARLS respondents aged 60–90. All specifications include the baseline controls and wave fixed effects. Standard errors clustered at the prefecture level in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

or sharing housework equally.

The behavioral and attitudinal evidence thus points in the same direction. Housework and grandparental care measure actual time allocation, whereas the CGSS items capture stated views about specialization between market and family roles. Because the surveys do not follow the same respondents, this agreement is convergence across datasets, not evidence that individual attitudes cause individual behavior.

5.3 Social Activities and Lifestyles

Health behavior offers an alternative explanation for the baseline pattern: clan culture could differentially alter women's and men's social engagement, exercise, sleep, drinking, or smoking (Ren et al., 2025). Table 9 provides little support for this account. All six interaction estimates are close to zero relative to their standard errors, including those for exercise, sleep, drinking, and smoking.

The supporting evidence is therefore concentrated in the organization of work and care rather than in the measured lifestyle outcomes. Women work less intensively in the labor market as clan intensity rises, while older men show a larger increase in grandparental childcare. These results narrow the set of plausible interpretations, but the data do not reveal the complete allocation of time over respondents' lives and cannot identify a causal mediation pathway.

6 Heterogeneity and Other Health Outcomes

6.1 Heterogeneity by Age and Residence

Age and rural residence capture two sources of variation in functional health and exposure to family organization. Disability rises sharply with age, while occupational histories, access to care, and reliance on family support differ between urban and rural China (Chen et al., 2018, 2022a; Gong et al., 2022; Murtagh and Hubert, 2004). I therefore compare respondents aged 45–59 with the baseline sample aged 60–90 and estimate the same specification separately for urban and rural residents. The mean incidence of any ADL limitation is 10.2% in the younger group and 23.4% in the older group, making the age comparison informative about whether the gender-differentiated association is already visible before old age.

The relationship is concentrated among older respondents. Column (1) of Table 10 estimates a gender difference in the clan gradient of 0.007 for respondents aged 45–59, indicating no detectable differential association in this group. Column (2) reproduces the baseline estimate of -0.044 among respondents aged 60–90. Column (3) pools both groups and instruments the full set of clan-age-gender interactions. The estimated gender differential is 5.0 percentage points more negative

Table 8. Gender-Role Attitudes

	Men: career; women: family	Men more capable	Marriage over work	Lay off women first	Share housework equally
	(1)	(2)	(3)	(4)	(5)
Panel A: All respondents					
Clan	0.159***	0.089*	0.101**	0.041	-0.053
	(0.056)	(0.047)	(0.046)	(0.041)	(0.045)
Obs.	11,347	11,334	11,317	11,171	11,341
Adj. R^2	0.051	0.038	0.029	0.030	0.011
KP rk F	25.26	25.25	25.19	25.10	25.20
Panel B: Women					
Clan	0.157***	0.080*	0.108*	0.056	-0.041
	(0.055)	(0.048)	(0.064)	(0.049)	(0.049)
Obs.	5,465	5,462	5,444	5,358	5,465
Adj. R^2	0.056	0.048	0.037	0.031	0.010
KP rk F	25.85	25.93	25.82	25.53	25.88
Panel C: Men					
Clan	0.163**	0.100*	0.102**	0.026	-0.062
	(0.064)	(0.061)	(0.044)	(0.048)	(0.052)
Obs.	5,882	5,872	5,873	5,813	5,876
Adj. R^2	0.045	0.030	0.022	0.030	0.015
KP rk F	24.63	24.54	24.51	24.57	24.49
Controls	YES	YES	YES	YES	YES
Wave FE	YES	YES	YES	YES	YES

Note: Each dependent variable records agreement with the statement shown in the column heading and ranges from 1 (strongly disagree) to 5 (strongly agree). Panel A reports estimates for all Han respondents aged 60–90; Panels B and C report estimates separately for women and men in the same age range. Academy distance instruments Clan. The data are from the 2010, 2012, 2013, 2014, and 2015 Chinese General Social Survey. All specifications include the baseline controls and survey-wave fixed effects. Standard errors clustered at the prefecture level in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 9. Social Activities and Lifestyles

	Social	Exercise	Log sleep time	Nap	Drink	Smoke
	(1)	(2)	(3)	(4)	(5)	(6)
Clan × Female	-0.006	0.001	0.022	-0.013	0.017	-0.001
	(0.013)	(0.008)	(0.014)	(0.016)	(0.016)	(0.019)
Controls	YES	YES	YES	YES	YES	YES
Wave FE	YES	YES	YES	YES	YES	YES
Obs.	26,077	15,976	24,831	25,144	26,062	24,714
Adj. R^2	0.015	0.034	0.030	0.030	0.172	0.257
KP rk F	29.47	29.99	29.98	29.72	29.45	29.63

Note: Social activity, exercise, napping, drinking, and smoking are indicator variables; sleep time is the natural logarithm of nightly sleep hours. The table reports Clan × Female; the main effects are included in estimation but omitted. Academy distance and its interaction with Female instrument Clan and Clan × Female. All specifications include the baseline controls and wave fixed effects. Standard errors clustered at the prefecture level in parentheses. Data are from CHARLS. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

among older respondents than among the younger group.

Columns (4)–(6) of Table 10 examine rural residence. The estimated female–male difference in the clan gradient is negative in both urban and rural samples, with a larger absolute value among urban residents. The pooled rural interaction is positive but does not differ detectably from zero. The data therefore do not support a clear urban-rural difference in the gender-differentiated association. This result is useful because clans were historically rooted in local and often rural social organization, yet lineage norms and kinship networks can persist after urbanization and migration (Greif and Tabellini, 2017; Peng, 2004).

The age contrast does not by itself identify a cumulative health effect. Current age also separates cohorts exposed to different fertility policies, labor-market conditions, and family institutions (Chen and Fang, 2021; Zhang, 2017). Survival and survey participation can further change the composition of the older sample (Banks et al., 2011). The estimates therefore show that the main pattern is specific to the older age range used in the baseline analysis; they cannot distinguish life-cycle accumulation from cohort differences or selective survival.

6.2 Other Dimensions of Healthy Aging

Healthy aging is multidimensional. Functional limitations capture whether health problems interfere with independent living, whereas cognitive ability and psychological well-being reflect different stocks of human capital and different responses to family and social environments (Chen et al., 2022a). The dimensions need not move together: an institution that reallocates physical

Table 10. Heterogeneity by Age and Rural Residence

	Age			Rural Residence		
	Aged 45–59	Aged 60–90	Diff.	Urban	Rural	Diff.
	(1)	(2)	(3)	(4)	(5)	(6)
Clan × Female	0.007	-0.044***	0.007	-0.053***	-0.036**	-0.053***
	(0.009)	(0.012)	(0.009)	(0.012)	(0.016)	(0.012)
Clan × Female × Aged 60–90			-0.050***			
			(0.012)			
Clan × Female × Rural						0.017
						(0.019)
Controls	YES	YES	YES	YES	YES	YES
Wave FE	YES	YES	YES	YES	YES	YES
Obs.	25,434	26,077	51,511	9,944	16,133	26,077
Adj. R^2	0.026	0.053	0.072	0.046	0.054	0.054
KP rk F	29.61	29.47	14.51	28.54	18.46	9.50

Note: The dependent variable is the indicator for any ADL limitation. Columns (1) and (2) estimate the baseline 2SLS specification separately for person-wave observations aged 45–59 and 60–90; column (3) pools both groups. Columns (4)–(6) analogously compare urban and rural residents. In each difference column, Clan, Clan × Female, Clan × Group, and Clan × Female × Group are instrumented by academy distance and the corresponding interactions. The first Clan × Female estimate in a difference column is the reference-group coefficient. Age is measured at each interview, so a respondent can contribute observations to different age groups after crossing age 60. All specifications include the baseline controls and wave fixed effects. Standard errors clustered at the prefecture level in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

work may affect ADLs without changing memory, while family companionship and support may influence emotional well-being more directly (Fang et al., 2023; Zhang, 2019). I therefore examine cognition, episodic memory, depressive symptoms, and life satisfaction. These are additional outcomes rather than mediators of the ADL result.

The cognition score ranges from 0 to 21 and episodic memory from 0 to 10; higher values indicate better cognitive function. The ten-item Center for Epidemiologic Studies Depression scale (CES-D) ranges from 0 to 30, with higher values indicating more depressive symptoms.[13] Life satisfaction ranges from 1 (not at all satisfied) to 5 (completely satisfied).

Table 11. Cognition and Psychological Well-Being

	Cognition (1)	Episodic memory (2)	CES-D (3)	Life satisfaction (4)
Clan × Female	-0.164	-0.053	-0.749***	0.047**
	(0.144)	(0.050)	(0.198)	(0.020)
Controls	YES	YES	YES	YES
Wave FE	YES	YES	YES	YES
Obs.	17,825	21,636	24,207	23,760
Adj. R^2	0.130	0.100	0.067	0.032
KP rk F	30.24	30.67	30.60	30.23

Note: The dependent variables measure cognitive ability, memory, depressive symptoms (CES-D), and self-rated life satisfaction. The table reports the coefficient of interest, Clan × Female; the main effects are included in estimation but omitted. All specifications include the baseline controls and wave fixed effects. Standard errors clustered at the prefecture level in parentheses. Data are from CHARLS. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

The cognitive outcomes do not reproduce the baseline pattern. The estimated female–male differences in the clan gradient are -0.164 for overall cognition and -0.053 for episodic memory, and neither differs statistically from zero. The psychological outcomes differ: a one-unit increase in Clan is associated with a 0.749-point reduction in women's CES-D score relative to men's and a 0.047-point relative increase in life satisfaction. Clan intensity is therefore not associated with a uniform female advantage across all dimensions of aging.

One possible interpretation is that clan-based family support and companionship are more closely connected to emotional well-being than to cognitive performance. This reading is compatible with evidence that family presence and intergenerational support matter for older people's welfare (Fang et al., 2023; Zhang, 2019), but the current estimates cannot isolate that pathway. CES-D, life satisfaction, and ADLs are measured contemporaneously, and unobserved health may affect both family interaction and reported well-being. The results therefore extend the outcome

[13] Following Andresen et al. (1994), the eight negatively worded items are scored from 0 for "rarely or none of the time" to 3 for "most or all of the time." The two positively worded items are reverse-coded.

profile and discipline the interpretation of the baseline coefficient; they do not establish that improved mental health causes better physical functioning.

7 Conclusion

This paper asks how an institution combining family insurance with patriarchal obligations shapes the gender gap in later-life health. Linking CHARLS respondents to prefecture-level genealogy records, I find that a one-standard-deviation increase in clan intensity narrows women's disadvantage in any ADL limitation by 3.8 percentage points, or 16% of the outcome mean. The pattern is robust across specifications and concentrated among respondents aged 60–90. It identifies a compression of women's disadvantage, not a reversal of the gender gap or a standalone improvement for women.

Supporting analyses associate stronger clan culture with higher fertility and more sons, lower relative market-work intensity among women, and a larger grandparental-childcare response among older men. Psychological well-being also improves for women relative to men, whereas cognition and measured lifestyles show little corresponding difference. These outcomes point to gendered life-course margins but do not identify mediation.

The design bounds this interpretation. Genealogy density measures a historical prefecture environment rather than individual membership, current residence may differ from lifetime exposure, and the panel excludes health trajectories after death or permanent attrition. Academy distance reduces some endogeneity concerns, but its exclusion restriction remains untestable. Subject to these qualifications, family institutions distribute both protection and cumulative burdens across gender.

References

Alesina, A., Giuliano, P., 2015. Culture and institutions. Journal of economic literature 53, 898–944.

Alesina, A., Giuliano, P., Nunn, N., 2013. On the origins of gender roles: Women and the plough. The quarterly journal of economics 128, 469–530.

Andresen, E.M., Malmgren, J.A., Carter, W.B., Patrick, D.L., 1994. Screening for depression in well older adults: evaluation of a short form of the ces-d. Prev Med 10, 77–84.

Banks, J., Muriel, A., Smith, J.P., 2011. Attrition and health in ageing studies: Evidence from elsa and hrs. Longitudinal and Life Course Studies 2, 101–126.

Basu, D., De Jong, R., 2010. Son targeting fertility behavior: Some consequences and determinants. Demography 47, 521–536.

Bau, N., 2021. Can policy change culture? government pension plans and traditional kinship practices. American Economic Review 111, 1880–1917.

Cao, J., Xu, Y., Zhang, C., 2022. Clans and calamity: How social capital saved lives during china's great famine. Journal of Development Economics 157, 102865.

Chen, X., Giles, J., Wang, Y., Zhao, Y., 2018. Gender patterns of eldercare in china. Feminist Economics 24, 54–76.

Chen, X., Giles, J., Yao, Y., Yip, W., Meng, Q., Berkman, L., Chen, H., Chen, X., Feng, J., Feng, Z., et al., 2022a. The path to healthy ageing in china: a peking university–lancet commission. The Lancet 400, 1967–2006.

Chen, Y., Fang, H., 2021. The long-term consequences of china's "later, longer, fewer" campaign in old age. Journal of Development Economics 151, 102664.

Chen, Z., Ma, C., Sinclair, A.J., 2022b. Banking on the confucian clan: why china developed financial markets so late. The Economic Journal 132, 1378–1413.

Ding, H., Li, C., Lu, X., Wang, H., 2025. Clan culture and supply chain resilience in china. Journal of Economic Behavior & Organization 238, 107219.

Ebrey, P.B., 2014. Confucianism and family rituals in imperial China: A social history of writing about rites. Princeton University Press.

Ervasti, J., Pentti, J., Nyberg, S.T., Shipley, M.J., Leineweber, C., Sørensen, J.K., Alfredsson, L., Bjorner, J.B., Borritz, M., Burr, H., et al., 2021. Long working hours and risk of 50 health conditions and mortality outcomes: a multicohort study in four european countries. The Lancet Regional Health–Europe 11.

Fan, H., Li, C., Xue, C., Yu, M., 2023. Clan culture and patterns of industrial specialization in china. Journal of Economic Behavior & Organization 207, 457–478.

Fang, H., Lei, Z., Lin, L., Zhang, P., Zhou, M., 2023. Family companionship and elderly suicide: Evidence from the chinese lunar new year. Journal of Development Economics 162, 103055.

Fei, H.T., 1946. Peasantry and gentry: An interpretation of chinese social structure and its changes. American Journal of Sociology 52, 1–17.

Fernández, R., Fogli, A., 2009. Culture: An empirical investigation of beliefs, work, and fertility. American economic journal: Macroeconomics 1, 146–177.

Fitzgerald, J., Gottschalk, P., Moffitt, R., 1998. An analysis of sample attrition in panel data: The michigan panel study of income dynamics. Journal of Human Resources 33, 251–299.

Freedman, M., 2021. Lineage organisation in South-eastern China. Routledge.

Gong, J., Wang, G., Wang, Y., Chen, X., Chen, Y., Meng, Q., Yang, P., Yao, Y., Zhao, Y., 2022. Nowcasting and forecasting the care needs of the older population in china: analysis of data from the china health and retirement longitudinal study (charls). The Lancet Public Health 7, e1005–e1013.

Greif, A., Tabellini, G., 2010. Cultural and institutional bifurcation: China and europe compared. American economic review 100, 135–140.

Greif, A., Tabellini, G., 2017. The clan and the corporation: Sustaining cooperation in china and europe. Journal of Comparative Economics 45, 1–35.

Gruber, J., Lin, M., Yi, J., 2023. The largest insurance expansion in history: saving one million lives per year in china. Technical Report. National Bureau of Economic Research.

Guiso, L., Sapienza, P., Zingales, L., 2006. Does culture affect economic outcomes? Journal of Economic perspectives 20, 23–48.

Guiso, L., Sapienza, P., Zingales, L., 2016. Long-term persistence. Journal of the European Economic Association 14, 1401–1436.

Guo, N., Huang, W., Wang, R., 2025. Public pensions and family dynamics: Eldercare, child investment, and son preference in rural china. Journal of Development Economics 172, 103390.

Huang, W., Zhang, C., 2021. The power of social pensions: evidence from china's new rural pension scheme. American Economic Journal: Applied Economics 13, 179–205.

Huang, Z., Jiang, X., Sun, A., 2024. Fertility and delayed migration: How son preference protects young girls against mother–child separation. Journal of Development Economics 166, 103191.

Kleibergen, F., Paap, R., 2006. Generalized reduced rank tests using the singular value decomposition. Journal of econometrics 133, 97–126.

Lei, X., Bai, C., Hong, J., Liu, H., 2022. Long-term care insurance and the well-being of older adults and their families: Evidence from china. Social Science & Medicine 296, 114745.

Mommaerts, C., 2018. Are coresidence and nursing homes substitutes? evidence from medicaid spend-down provisions. Journal of health economics 59, 125–138.

Murtagh, K.N., Hubert, H.B., 2004. Gender differences in physical disability among an elderly cohort. American journal of public health 94, 1406–1411.

Nathanson, C.A., 1975. Illness and the feminine role: a theoretical review. Social Science & Medicine (1967) 9, 57–62.

Oliveira, J., 2016. The value of children: Inter-generational support, fertility, and human capital. Journal of Development Economics 120, 1–16.

Peng, Y., 2004. Kinship networks and entrepreneurs in china's transitional economy. American Journal of Sociology 109, 1045–1074.

Ren, L., Zhou, Y., Liu, K., Zhang, H., Li, S., Hu, Y., Shirai, K., Jiang, Y., Wu, Y., Yu, M., et al., 2025. The effect of healthy lifestyles and social determinants on independent life expectancy and sex differences in china: evidence from a 13-year cohort study. The Lancet Public Health 10,

e1016–e1024.

Shiue, C.H., 2017. Human capital and fertility in chinese clans before modern growth. Journal of Economic Growth 22, 351–396.

Spector, W.D., Fleishman, J.A., 1998. Combining activities of daily living with instrumental activities of daily living to measure functional disability. The Journals of Gerontology Series B: Psychological Sciences and Social Sciences 53, S46–S57.

Sun, A., Xia, F., Zhang, X., 2025. The motherhood penalty on health: Evidence from china. Journal of Economic Behavior & Organization 238, 107241.

Szonyi, M., 2002. Practicing Kinship: Lineage and Descent in Late Imperial China. Stanford University Press.

Talhelm, T., Zhang, X., Oishi, S., Shimin, C., Duan, D., Lan, X., Kitayama, S., 2014. Large-scale psychological differences within china explained by rice versus wheat agriculture. Science 344, 603–608.

Tang, C., Zhao, Z., 2023. Informal institution meets child development: Clan culture and child labor in china. Journal of Comparative Economics 51, 277–294.

Wang, J., Lao, W., Qin, F., 2026. Clan culture and e-commerce development. The Journal of World Economy 49, 111–142.

Zhang, C., 2019. Family support or social support? the role of clan culture. Journal of Population Economics 32, 529–549.

Zhang, C., Li, T., 2017. Culture, fertility and the socioeconomic status of women. China Economic Review 45, 279–288.

Zhang, J., 2017. The evolution of china's one-child policy and its effects on family outcomes. Journal of Economic Perspectives 31, 141–160.

Zhang, L., 2022. Patrilineality, fertility, and women's income: Evidence from family lineage in china. China Economic Review 74, 101805.

Zhao, Y., Strauss, J., Yang, G., Giles, J., Hu, P., Hu, Y., Lei, X., Park, A., Smith, J.P., Wang, Y., 2013. China health and retirement longitudinal study–2011–2012 national baseline users' guide. Beijing: National School of Development, Peking University 2, 1–56.

Zheng, Z., 2001. Family Lineage Organization and Social Change in Ming and Qing Fujian. University of Hawai'i Press.

Appendix

A. Figures

The following figures supplement the main descriptive evidence and robustness analysis.

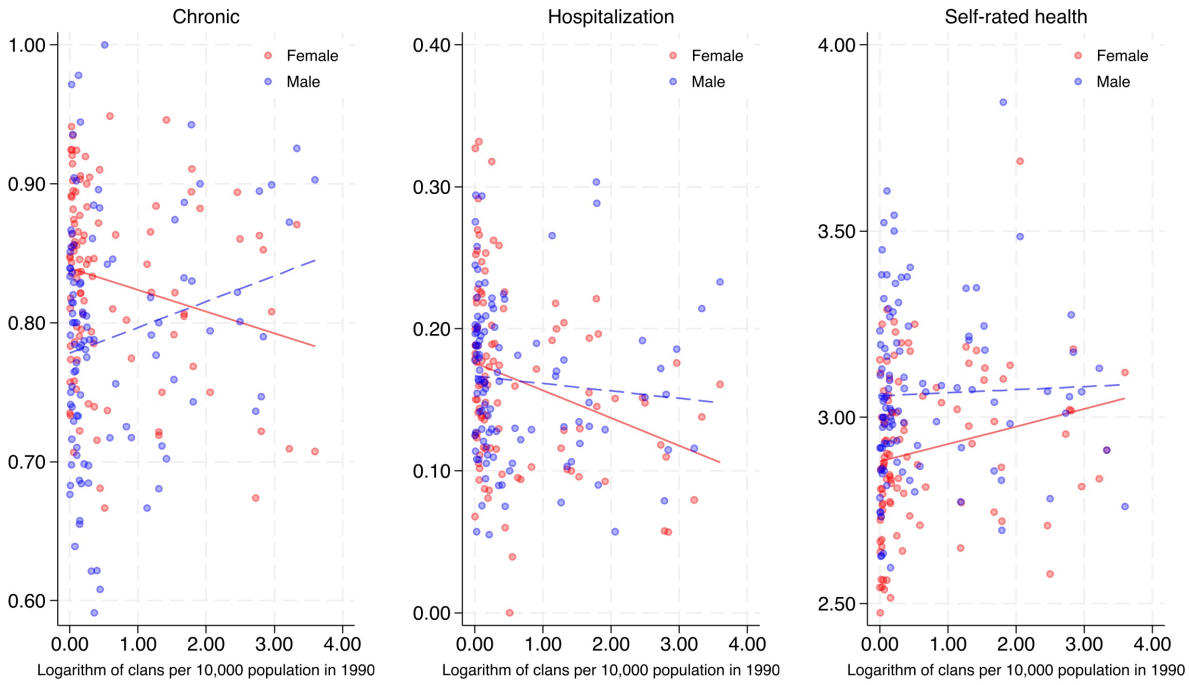


Figure A.1. Clan Intensity and Alternative Health Measures

Note: The figure plots prefecture-level mean chronic conditions, hospitalization, and self-rated health against clan intensity separately for women and men. The solid line represents women and the dashed line represents men. Clan intensity is the natural logarithm of one plus genealogy books compiled before 1990 per 10,000 residents.

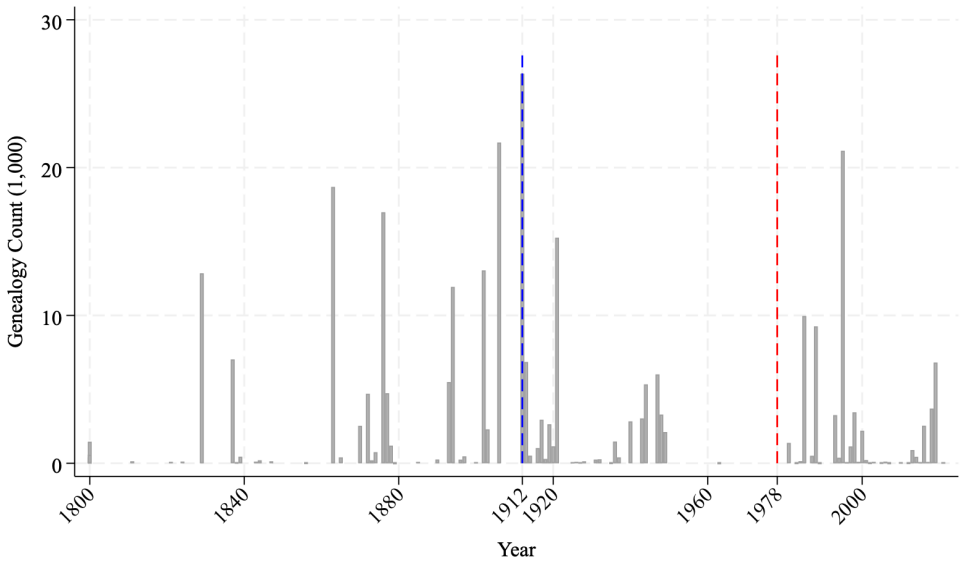


Figure A.2. Number of Genealogies Across Historical Periods

Note: This figure shows recorded genealogy compilations since 1800. Compilation was more common during the late Qing dynasty and after the reform and opening-up period, and was rare around the Cultural Revolution.

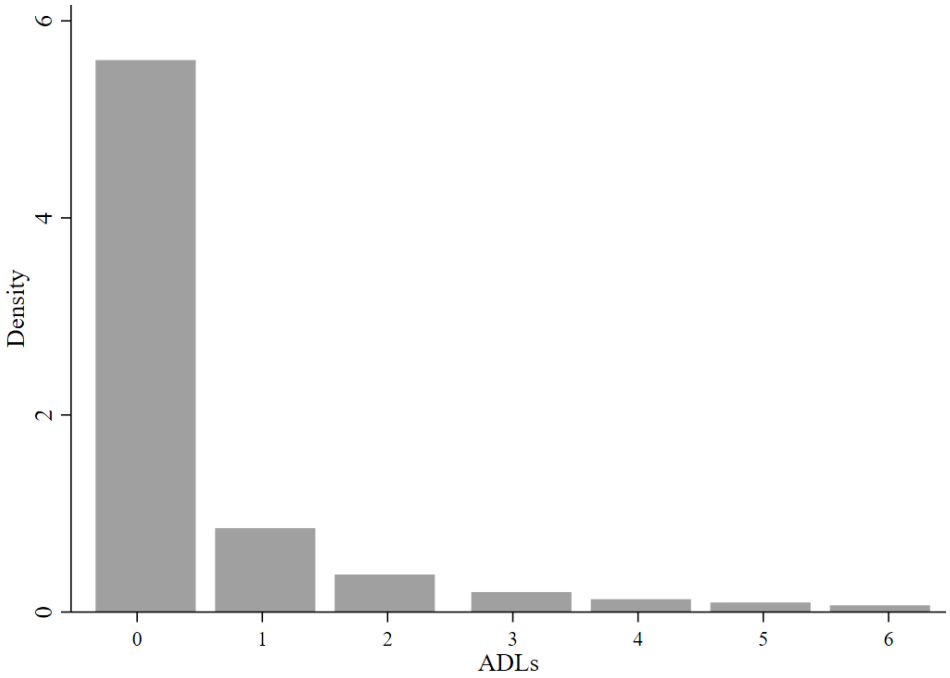


Figure A.3. Distribution of ADL Limitations

Note: The ADL score ranges from 0 to 6 and has a large mass at zero. This distribution motivates the binary indicator used in the baseline analysis.

B. Tables

The following tables document sample construction, data sources, identification and measurement checks, and supplementary mechanism evidence.

Table B.1. Sample Construction and Panel Coverage

Panel A: Sample construction	
All CHARLS person-wave observations	96,628
2011–2018 observations	77,233
Matched to prefecture covariates and academy distance	71,940
Han respondents aged 60–90	28,348
Baseline complete-case observations	26,077
Panel B: Panel coverage and outcome availability	
Unique respondents	9,092
Unique households	5,954
Unique prefectures	105
2011 / 2013 / 2015 / 2018 observations	4,377 / 5,634 / 7,059 / 9,007
Respondents observed in one / two / three / four waves	1,754 / 1,624 / 1,781 / 3,933
ADL observed among eligible men	13,772 / 13,860 (99.37%)
ADL observed among eligible women	14,384 / 14,488 (99.28%)
Next-wave ADL observed, men	8,054 / 8,293 (97.12%)
Next-wave ADL observed, women	8,510 / 8,777 (96.96%)

Note: The baseline sample includes Han CHARLS respondents aged 60–90 in 2011–2018 with nonmissing ADL limitation, age, rural status, and prefecture-level controls. Observations are matched to the prefecture-level genealogy, geographic-control, and academy-distance data. The panel is unbalanced; 473 respondents (5.2%) have a gap between observed survey waves. Next-wave ADL availability is measured among complete baseline observations in 2011–2015, without imposing the next wave's age or complete-case restrictions. The processed file contains only four recorded death flags and no proxy-interview indicator, so it cannot separate mortality from other nonresponse.

Table B.2. Data Sources

Variables	Data source and years
Alternative health, fertility, employment, childcare, cognition, mental health, social activity, and lifestyle	China Health and Retirement Longitudinal Study (CHARLS), 2011, 2013, 2015, and 2018.
Religious-site controls	National Religious Affairs Administration registry of religious activity venues; Buddhist and Taoist sites are geocoded and aggregated to prefectures.
Housework time	Chinese General Social Survey (CGSS), 2015.
Gender-role attitudes	Chinese General Social Survey (CGSS), 2010, 2012, 2013, 2014, and 2015.
Children, sons, and son ratio	1990 Population Census.
GDP, grain output, secondary schools, and physicians	*China City Statistical Yearbook*, 1990.
Genealogy records	Shanghai Library, *General Catalog of Chinese Genealogy*; records compiled before 1990. https://jiapu.library.sh.cn/
Academy distance	Locations of academies at which Zhu Xi taught, compiled by Chen et al. (2022b).
Geographic controls	Project GIS layers for coastlines, Qing treaty ports, rivers, prefecture boundaries, and terrain.

Note: This table documents the data sources and survey years for variables used in the supplementary analyses. Individual observations are linked to prefecture-level variables using prefecture of residence.

Table B.3. Academy Distance and 1990 Prefecture Characteristics

	GDP	Grain output	Secondary schools	Physicians
	(1)	(2)	(3)	(4)
Log academy distance	0.227	0.140	0.114	0.181
	(0.184)	(0.215)	(0.166)	(0.158)
Geographic controls	YES	YES	YES	YES
Prefectures	94	94	94	94
Adj. R^2	0.192	0.017	0.028	0.084

Note: This table reports associations, not a test of the exclusion restriction. The 1990 *China City Statistical Yearbook* variables are unavailable for 11 of the 105 prefectures in the baseline sample, so all four columns use the remaining 94 prefectures. The outcomes are the natural logarithms of prefecture-level total GDP (ten thousand yuan), grain output (ten thousand tons), the number of secondary schools, and the number of physicians measured in 1990. All specifications include the baseline geographic controls. Standard errors clustered at the prefecture level in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table B.4. Son Preference

	Number of children	Number of sons	Son share
	(1)	(2)	(3)
Clan	0.042***	0.024***	0.002**
	(0.014)	(0.008)	(0.001)
Controls	YES	YES	YES
Obs.	3,615,770	3,615,770	1,254,834
Adj. R^2	0.009	0.008	0.000
KP rk F	80.59	80.59	85.38

Note: The data are from the 1990 Population Census. Columns (1) and (2) report the numbers of children and sons; column (3) reports the share of children who are sons. Academy distance instruments Clan. All specifications include rural residence and the baseline geographic controls. Standard errors clustered at the prefecture level in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table B.5. Gender Differences in Housework Time

	Weekday minutes		Weekend minutes	
	Self-reported	Spouse-reported	Self-reported	Spouse-reported
	(1)	(2)	(3)	(4)
Clan	13.357	44.149**	4.193	40.788**
	(14.965)	(17.356)	(11.971)	(16.297)
Female	116.375***	-37.606**	114.487***	-33.817
	(13.309)	(19.024)	(15.964)	(20.759)
Clan × Female	8.345	-56.769**	21.570	-71.958***
	(18.543)	(24.227)	(18.547)	(24.625)
Controls	YES	YES	YES	YES
Obs.	377	331	380	330
Adj. R^2	0.250	0.077	0.253	0.091
KP rk F	12.05	10.51	10.53	9.84

Note: The dependent variable is daily housework time in minutes. The sample consists of Han respondents aged 45–59 in the 2015 Chinese General Social Survey. Columns (1) and (3) use respondents' reports of their own housework time; columns (2) and (4) use their reports of their spouse's housework time. In the spouse-reported columns, the interaction compares the clan gradient in husbands' time, reported by women, with the gradient in wives' time, reported by men. The negative interactions imply that the wife-minus-husband gap rises by 56.8 weekday minutes and 72.0 weekend minutes for a one-unit increase in Clan. The self-reported interactions are positive and do not differ detectably from zero. All specifications include age, rural residence, distance to the coastline and port, river density, and slope. Standard errors clustered at the prefecture level in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.